Counting composite prime powers in consecutive prime gaps

A rigorous related result for OEIS A080101

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Abstract

OEIS A080101 counts composite prime powers strictly between consecutive primes. Its conjectured universal maximum of two remains unproved. We establish an exact formula for the cumulative values, deduce from the prime number theorem an asymptotic of order $\sqrt{N/\log N}$, and show that the sequence is positive infinitely often but is zero at a set of indices of natural density one. We also record a direct restriction on prime-power bases inside a prime gap. None of these statements proves the conjectured pointwise bound.

Precise OEIS assertion and scope

Let p_n be the nth prime ($p_1 = 2$). Following OEIS A080101 [1], put

$$a(n) = \#\{q^k : p_n < q^k < p_{n+1},\ q \text{ prime},\ k \geq 2\}.$$

The entry records that the values $0, 1, 2$ occur and conjectures that $a(n) \leq 2$ for every n. This conjecture is *not* assumed below, and the results here do not establish it.

Define the cumulative count $S(N) = \sum_{n=1}^{N} a(n)$ and $X_N = p_{N+1} - 1$. Write $\pi(x)$ for the number of primes $\leq x$ and log for the natural logarithm.

Theorem 1 (Exact summatory formula and asymptotic)**.** *For every integer $N \geq 1$,*

$$S(N) = \sum_{k=2}^{\lfloor \log_2 X_N \rfloor} \pi\left(\left\lfloor X_N^{1/k} \right\rfloor\right), \tag{1}$$

where an empty sum means zero. Moreover, as $N \to \infty$,

$$S(N) \sim \frac{2\sqrt{p_{N+1}}}{\log p_{N+1}} \sim 2\sqrt{\frac{N}{\log N}}. \tag{2}$$

Proof. Every composite prime power q^k, $k \geq 2$, is strictly between two unique consecutive primes. Thus the union of the first N prime gaps contains exactly the composite prime powers $q^k \leq X_N$. Their representation with prime base q and exponent $k \geq 2$ is unique by prime factorization. For fixed k, the allowable bases are precisely the primes $q \leq X_N^{1/k}$, giving (1).

For $X \to \infty$, write $Q(X) = \sum_{k=2}^{\lfloor \log_2 X \rfloor} \pi(\lfloor X^{1/k} \rfloor)$. The terms with $k \geq 3$ contribute at most

$$\sum_{k=3}^{\lfloor \log_2 X \rfloor} X^{1/k} \leq X^{1/3} \log_2 X = o\left(\frac{\sqrt{X}}{\log X}\right).$$

Hence $Q(X) = \pi(\lfloor \sqrt{X} \rfloor) + o(\sqrt{X}/\log X)$. The prime number theorem [2] gives

$$\pi(\lfloor \sqrt{X} \rfloor) \sim \frac{\sqrt{X}}{\log \sqrt{X}} = \frac{2\sqrt{X}}{\log X}.$$

Taking $X = X_N$, and using $p_{N+1} \sim N \log N$ and $\log p_{N+1} \sim \log N$ (also consequences of the prime number theorem), proves (2). □

Corollary 1 (Infinitely many nonzero terms, but density zero)**.** *There are infinitely many n with $a(n) > 0$. Nevertheless,*

$$\#\{n \le N : a(n) > 0\} \le S(N) = (2 + o(1))\sqrt{N/\log N} = o(N).$$

Thus $a(n) = 0$ for a set of indices of natural density one, and $N^{-1}\sum_{n\le N} a(n) \to 0$.

Proof. Every square q^2 of a prime is a composite prime power between consecutive primes. Since there are infinitely many such squares and each finite prime gap contains finitely many integers, infinitely many different gaps have $a(n) > 0$. The bound follows from $\mathbf{1}_{\{a(n)>0\}} \le a(n)$ and Theorem 1. □

Lemma 1 (Distinct prime bases inside a single gap)**.** *A fixed prime q contributes at most one composite power q^k to any particular interval (p_n, p_{n+1}).*

Proof. If $q^r < q^s$ with $2 \le r < s$, then $q^s \ge 2q^r$. Bertrand's postulate supplies a prime strictly between q^r and $2q^r$, hence between q^r and q^s. The two powers cannot lie in the same prime gap. □

Checks and limitations

For example, $a(4) = 2$ because $7 < 8 = 2^3 < 9 = 3^2 < 11$. OEIS also records $a(327) = 2$, from $2187 = 3^7$ and $2197 = 13^3$ between 2179 and 2203 [1]. Direct enumeration through $N = 10{,}000$ matches (1) exactly; these finite checks are not used in the proof. The asymptotic is a *summatory* statement: arbitrarily rare values $a(n) \ge 3$ would not contradict it. In particular, no uniform upper bound $a(n) \le 2$ follows.

Priority and authorship note. The exact counting idea is closely related to formulas and code already recorded in OEIS A080101, while the asymptotic and density statements follow from standard prime-counting facts. No novelty or first-proof priority is claimed. This AI-authored exposition was algebraically and computationally checked during preparation but has not undergone independent peer review.

References

[1] The OEIS Foundation, "A080101: Number of prime powers in all composite numbers between n-th prime and next prime," https://oeis.org/A080101, accessed November 13, 2026.

[2] NIST Digital Library of Mathematical Functions, §27.12, "Asymptotic Formulas: Primes," https://dlmf.nist.gov/27.12.

[3] J. Bertrand's postulate (Chebyshev's theorem), the classical assertion that for every integer $m > 1$ there is a prime r with $m < r < 2m$.